

SOIC 14-Pin	Pin Name (1,2)	Other/Special	ADC0	AC0	DAC0	USART0	SPI0	TWI0	TCA0	TCB0	TCF0	CCL
10	PA0	RESET/UPDI	AIN0									LUT0-IN0
11	PA1		AIN1			TxD ⁽³⁾	MOSI	SDA ⁽³⁾				LUT0-IN1
12	PA2	EVOUT0	AIN2			RxD ⁽³⁾	MISO	SCL ⁽³⁾				LUT0-IN2
13	PA3	EXTCLK	AIN3			XCK ⁽³⁾	SCK		WO3			
14	GND											
1	VDD											
2	PA4		AIN4			XDIR ⁽³⁾	SS		WO4		WOA	LUT0-OUT
3	PA5		AIN5	OUT					WO5	WO	WOB	
4	PA6		AIN6	AINN0	OUT		MOSI ⁽³⁾					
5	PA7		AIN7	AINP0			MISO ⁽³⁾					LUT1-OUT
6	PB3	TOSC1				RxD			WO0 ⁽³⁾			
7	PB2	TOSC2, EVOUT1				TxD			WO2			
8	PB1		AIN10			XCK		SDA	WO1			
9	PB0		AIN11			XDIR		SCL	WO0			

Fig. 6: Datasheet table of MCU pin descriptions (Courtesy: Microchip – <https://ww1.microchip.com/downloads/en/DeviceDoc/40001911A.pdf>)

Wires were soldered directly to the required pins for this implementation.

Next, you will need to add support for the ATtiny series in the Arduino IDE. There are several

ways to do this. The steps are detailed in here, and you can also follow along using the reference video link:

Reference video tutorial. For a step-by-step visual reference, this tutorial provides a clear walk-through of programming the ATtiny412 MCU using an Arduino UNO: <https://www.youtube.com/watch?v=Idul7NfqfIk>

To enable support for

ATtiny series microcontrollers in the Arduino IDE, begin by adding the following board manager URL to the preferences section: http://drazzy.com/package_drazzy.com_index.json. Once added, install the megaTinyCore package through the Boards Manager. After successful installation, select ATtiny412 as the target board for programming. Fig. 4 shows the selection of the ATtiny412 MCU chip in the Arduino IDE for uploading/flashing the firmware. Refer link: https://www.youtube.com/watch?v=AL9vK_xMt4E

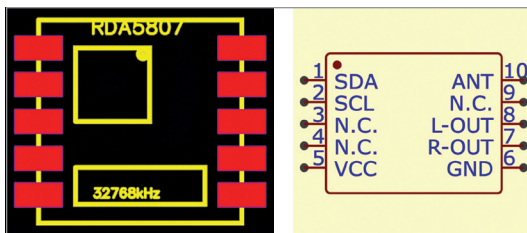


Fig. 7: RDA5807 module footprint

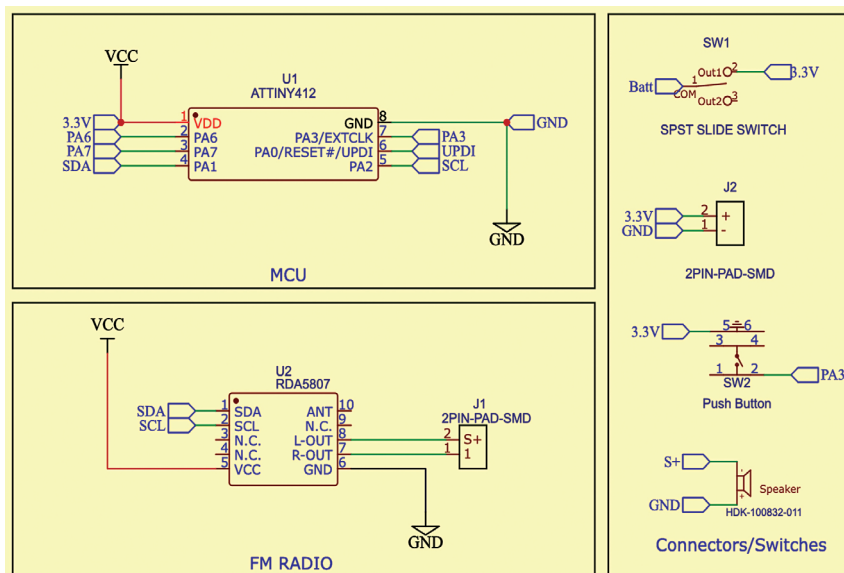


Fig. 8: Circuit diagram

Circuit

According to the ATtiny412 MCU datasheet, the pinout is shown in Fig. 5, and the pin descriptions are detailed in Fig. 6. The microcontroller operates at voltages as low as 3.3V. The I²C port uses pins PA1 (SDA) and PA2 (SCL). The complete datasheet of the MCU can be referred to at this link: <https://ww1.microchip.com/downloads/en/DeviceDoc/40001911A.pdf>

Fig. 7 displays the RDA5807 FM radio module's footprint. The module includes I²C interface pins, stereo audio output, and power supply connections.

Fig. 8 illustrates the complete